

# ELIA RÜHLE

elia.ruehle@gmail.com

**Research Interests:** Representation Learning, World Models, Embodied AI, Mechanistic Interpretability, Robotics

## EDUCATION

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### Technische Universität Dresden

October 2023 - November 2026

M.Sc. in Computer Science with focus on Artificial Intelligence - Current GPA: 1.1/1.0 (German scale)

Thesis: Leveraging Latent Space Structure in Predictive World Models for Long-Horizon Planning

Supervisor: Prof. Roberto Calandra, Prof. Christel Baier

### Technische Universität Dresden

October 2019 - September 2023

B.Sc. in Computer Science & Mathematics with graduation in Computer Science - GPA: 1.5/1.0 (German scale)

Thesis: Learning Grasp-Stability with Multi-Finger Hands and Touch in Simulation - 1.0/1.0 (German scale)

Supervisor: Prof. Roberto Calandra, Prof. Uwe Aßmann

## RESEARCH EXPERIENCE

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### TU Dresden - Learning, Adaptive Systems and Robotics (LASR) Lab

October 2023 - January 2026

*Research Assistant*

*Dresden*

- Investigated diffusion-based planners for curiosity-driven exploration in model-based reinforcement learning
- Developed a scalable simulation framework for omnidirectional tactile sensors, enabling large-scale parallel robotic manipulation experiments in Isaac Gym
- Designed and implemented simulation pipelines for learning-based grasping with multi-finger robotic hands
- Conducted a comprehensive literature review on tactile robotic grasping and developed a unified taxonomy for learning-based grasping approaches
- Supervised by Prof. Roberto Calandra and Johannes Busch

### TU Dresden - Chair of Machine Learning for Computer Vision

March 2023 - September 2023

*Research Assistant*

*Dresden*

- Conducted research at the intersection of discrete optimization and computer vision, focusing on the development and implementation of image segmentation algorithms
- Contributed to solving the multi-separator problem by designing and evaluating optimization-based algorithms using C++ and Python
- Supervised by Prof. Björn Andres

## TEACHING EXPERIENCE

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### TU Dresden - Faculty of Computer Science

March 2023 - September 2023

*Instructor*

*Dresden*

- Weekly tutorials (~15 participants) on foundational and advanced concepts of the Python programming language
- Prepared slides and programming assignments which were solved and demonstrated during interactive live sessions

### TU Dresden - Faculty of Mathematics

October 2020 - March 2022

*Teaching Assistant*

*Dresden*

- Teaching Assistant for Linear Algebra I & II (Mathematics program) and Discrete Structures (Computer Science program)
- Conducted weekly tutorials for groups of approximately 30 students, supported the design and grading of exercises, and assisted with the preparation and evaluation of written and oral examinations

SELECTED RESEARCH PROJECTS

**Join Order Selection using Cardinalities and Reinforcement Learning**  
*Database Research Group*

April 2025 - September 2025  
*Dresden*

- Investigated reinforcement learning-based query optimization by learning join ordering policies from cardinalities and query interactions.
- Analyzed the influence of hyperparameters and training data diversity on learning stability and generalization
- Achieved a 25.37 mean reward improvement during training and demonstrated improved generalization with larger and more diverse query workloads
- Supervised by Prof. Wolfgang Lehner

**Decoding Emotions: Learning Feelings from Textual Data**  
*ScaDS.AI*

October 2024 – March 2025  
*Dresden*

- Developed a transformer-based model for emotion prediction from textual data, achieving 76.57% F1-score in the SemEval 2025 competition
- Supervised by Prof. Michael Färber and Prof. Simon Razniewski

**Unveiling Active Learning Strategies: Similarities, Applicability to Unlabeled Data and Influence of Initial Performance**  
*Database Research Group*

April 2023 - August 2023  
*Dresden*

- Clustered 37 Active Learning (AL) strategies to identify similarities and support systematic strategy selection across diverse machine learning tasks
- Developed a decision tree-based recommendation framework for selecting effective AL strategies based on dataset characteristics, achieving 69.2% accuracy on unseen datasets
- Supervised by Prof. Wolfgang Lehner and Dr. Julius Gonsior

SKILLS & INTERESTS

|                              |                                                                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | Python, L <sup>A</sup> T <sub>E</sub> X, Bash, C/C++                                                                                         |
| <b>Tools</b>                 | PyTorch, NumPy, Scikit-Learn, Issac Gym, MuJoCo, JAX, Linux, Git                                                                             |
| <b>Languages</b>             | German(native), English(fluent), Latin                                                                                                       |
| <b>Interests</b>             | Representation Learning, World Models, Robot Learning, Embodied AI, Multimodal Learning, Physical Intelligence, Mechanistic Interpretability |

REFEREES

**Prof. Dr. Roberto Calandra**  
Machine Learning for Robotics (LASR Lab)  
TU Dresden, Germany  
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**Prof. Dr. Wolfgang Lehner**  
Database Research Group  
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